

Multivariate Analysis – Honours

Section 1 Exercises Solutions

Chapter 1

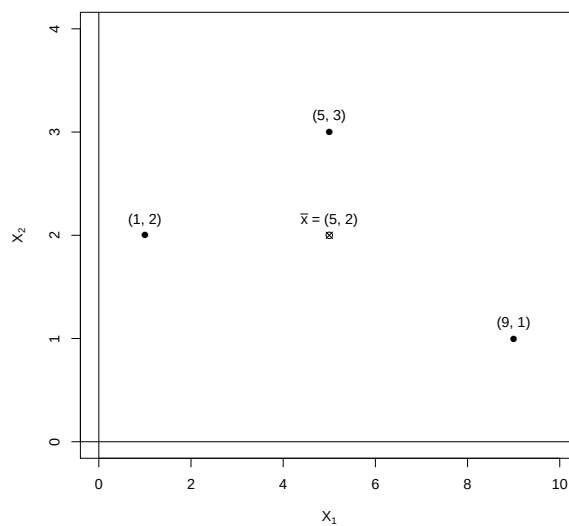
See the file `chapter_1_solutions.R` for the accompanying code.

Homework exercise 1.1

1.

$$\mathbf{X} = \begin{bmatrix} 9 & 1 \\ 5 & 3 \\ 1 & 2 \end{bmatrix}$$

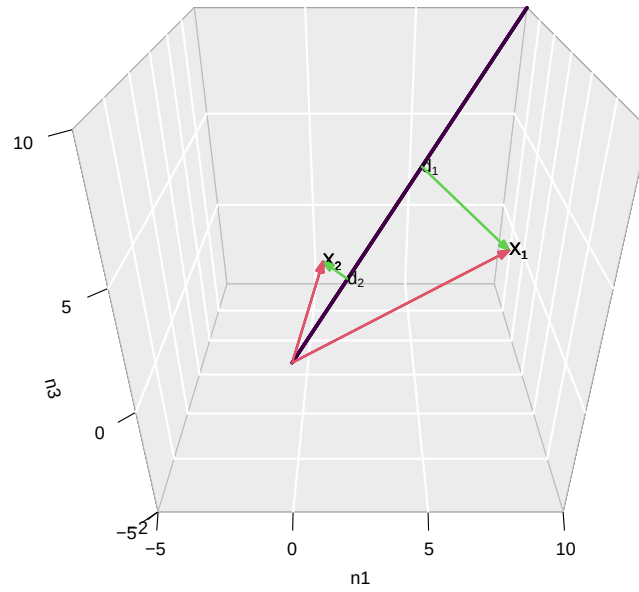
$$\bar{\mathbf{x}} = \frac{1}{3}\mathbf{X}'\mathbf{1} = \frac{1}{3} \begin{bmatrix} 9 & 5 & 1 \\ 1 & 3 & 2 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 15 \\ 6 \end{bmatrix} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$$



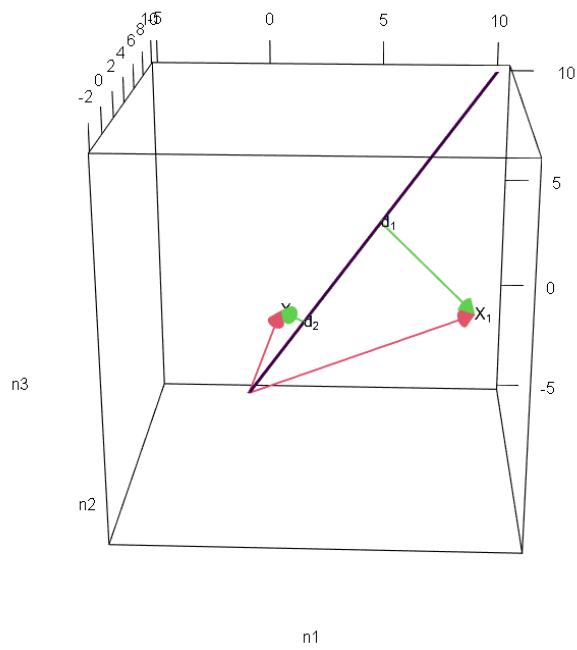
2.

$$\mathbf{d}_1 = \mathbf{x}_1 - \bar{x}_1 \mathbf{1} = \begin{bmatrix} 9 \\ 5 \\ 1 \end{bmatrix} - \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \\ -4 \end{bmatrix}$$

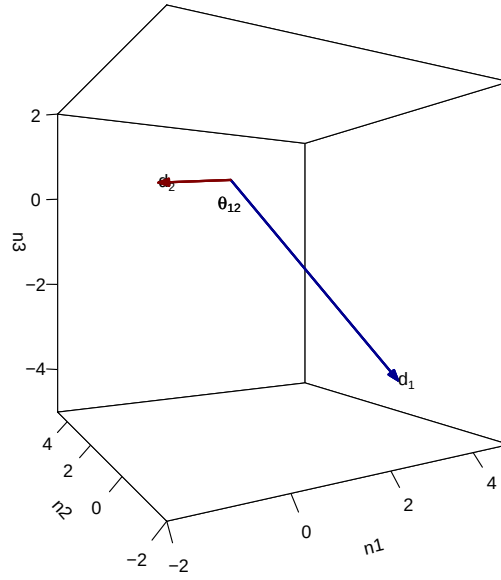
$$\mathbf{d}_2 = \mathbf{x}_2 - \bar{x}_2 \mathbf{1} = \begin{bmatrix} 1 \\ 3 \\ 2 \end{bmatrix} - \begin{bmatrix} 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}$$



Sometimes it's difficult to estimate the depth in 3D plots. Spinning helps!



3.



$$L_{d_1} = \sqrt{\mathbf{d}'_1 \mathbf{d}_1} = \sqrt{32}$$

$$L_{d_2} = \sqrt{\mathbf{d}'_2 \mathbf{d}_2} = \sqrt{2}$$

$$\mathbf{d}'_1 \mathbf{d}_2 = -4$$

$$\cos(\theta) = \frac{\mathbf{d}'_1 \mathbf{d}_2}{L_{d_1} L_{d_2}} = \frac{-4}{\sqrt{(32)(2)}} = -0.5$$

$$(n-1)s_{11} = L_{d_1}^2 \implies s_{11} = \frac{32}{2} = 16$$

$$(n-1)s_{22} = L_{d_2}^2 \implies s_{22} = \frac{2}{2} = 1$$

$$(n-1)s_{12} = \mathbf{d}'_2 \mathbf{d}_2 \implies s_{12} = \frac{-4}{2} = -2$$

$$r_{12} = \cos(\theta) = -0.5$$

$$\therefore \mathbf{S} = \begin{bmatrix} 16 & -2 \\ -2 & 1 \end{bmatrix} \quad \& \quad \mathbf{R} = \begin{bmatrix} 1 & -0.5 \\ -0.5 & 1 \end{bmatrix}$$

Homework exercise 1.2

1. x_1 = petroleum, x_2 = natural gas, x_3 = hydroelectric power and x_4 = nuclear electric power is recorded in quadrillions (10^{15}).

$$\bar{\mathbf{x}} = \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix}$$

and

$$\mathbf{S} = \begin{bmatrix} 0.856 & 0.635 & 0.173 & 0.096 \\ 0.635 & 0.568 & 0.128 & 0.067 \\ 0.173 & 0.128 & 0.171 & 0.039 \\ 0.096 & 0.067 & 0.039 & 0.043 \end{bmatrix}$$

Let Y = A state's total energy consumption for these resources.

$$\therefore Y = X_1 + X_2 + X_3 + X_4 = \mathbf{c}'\mathbf{X}$$

where $\mathbf{c}' = [1 \quad 1 \quad 1 \quad 1]$ and $\mathbf{X} = \begin{bmatrix} X_1 \\ X_2 \\ X_3 \\ X_4 \end{bmatrix}$

Then

$$\bar{y} = \mathbf{c}'\bar{\mathbf{x}} = [1 \quad 1 \quad 1 \quad 1] \begin{bmatrix} 0.766 \\ 0.508 \\ 0.438 \\ 0.161 \end{bmatrix} = 1.873$$

and

$$\mathbf{S}_Y = \mathbf{c}'\mathbf{S}\mathbf{c} = 3.914$$

2. Let Z = The excess of petroleum over natural gas consumption

$$\therefore Z = X_1 - X_2 = \mathbf{b}'\mathbf{X}$$

where $\mathbf{b}' = [1 \quad -1 \quad 0 \quad 0]$

Then

$$\bar{z} = \mathbf{b}'\bar{\mathbf{x}} = 0.766 - 0.508 = 0.258$$

and

$$\mathbf{S}_Z = \mathbf{b}'\mathbf{S}\mathbf{b} = 0.154$$

The sample covariance between Y and Z is given by $\text{Cov}(Y, Z) = \mathbf{b}'\mathbf{S}\mathbf{c} = 0.362$.
